

Electron energy loss spectroscopy and anion formation in gas phase coronene

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Electron collisions with gas phase coronene have been studied in the energy range 0–80 eV to investigate electronic and pure vibrational excitations, as well as the anion formation near zero energy. Quantum chemical calculations using time dependent density functional theory (TDDFT) have been performed for an improved data analysis. Results have been obtained on both dipole allowed and forbidden transitions to low lying electronic excited states up to 8 eV, by varying the incident energy and scattering observation angle. Calculated values for excitation energies are in good agreement with experiment and with previous data. Pure vibrational excitation, performed with an energy resolution of 0.025 eV, is presented at incident energies from 1.5 to 9 eV. Five groups of excited vibrational modes are observed with energy losses ranging from 0.065 to 0.375 eV. Several resonant states appear to contribute to these excitations. Calculations of the harmonic vibrational frequencies of coronene, and determination of their main character, are used for the analysis of the results. The vertical electron affinity of coronene is calculated to be 0.45 eV in very good agreement with previous measured or calculated results. The parent anion ($\text{C}_{24}\text{H}_{12}^-$) yield *versus* electron energy presents 2 peaks at zero and 0.30 eV (within 0.030 eV), in contrast with previous results where only one peak at zero was reported [Denifl *et al.*, *J. Chem. Phys.*, 2005, 123, 104308].

1. Introduction

Coronene $\text{C}_{24}\text{H}_{12}$ (hexabenzobenzene) is a large polycyclic aromatic hydrocarbon (PAH) molecule. It is planar and belongs to the family of pericondensed benzenoid aromatic molecules $\text{C}_{6n2}\text{H}_{6n}$ ($n = 2$) presenting a D_{6h} symmetry, like benzene ($n = 1$) and circumcoronene $\text{C}_{54}\text{H}_{18}$ ($n = 3$). Like other PAHs, it is considered to play an important role in the interstellar medium.^{1,2} Furthermore, due to its large number of carbon atoms inter-connected in 6-membered planar rings, coronene is often considered as a model molecule for studies on graphene surfaces.³ A very important amount of work concerning the spectroscopy of the molecule has already been published (see ref. 4 and references therein), however, only a few studies concern the behaviour of coronene under electron impact.^{5–9} Low energy (0–10 eV) electron collisions with molecules give rise to processes very different from photon–molecule interactions. These processes are dominated by the existence of transitory anion states (“resonances”) which are known to efficiently excite vibrationally or dissociate the molecule, even below the bond energy.¹⁰ This work, performed with an electron energy resolution of 0.025 eV, presents a detailed study of vibrational excitation in the energy range 1.5–9 eV. We also present a study of transitions, both dipole allowed and forbidden, to low lying electronic

states (up to 8 eV). The formation of the long lived $\text{C}_{24}\text{H}_{12}^-$ anion has been revisited. Pure vibrational excitation and electronic excitation experiments are interpreted in the light of quantum chemical calculations using the density functional theory.

2. Experimental

The set up used for the present experiment is an electrostatic electron spectrometer equipped with two hemispherical analysers in tandem, both in the electron gun and in the analyser sections. Electron optics and magnetic shielding have been carefully designed to allow both the electron gun and the analyser to work down to zero energy (within 0.030 eV). Energy resolution ranges from 0.025 eV to 0.050 eV (FWHM energy loss mode) with electron currents of 0.5 to 5 nA. Mass analysis of positive or negative ions is achieved by a time of flight spectrometer based on a McLaren–Wiley geometry, the ions being collected onto microchannel plates. For anion studies, the electron beam is pulsed off during the mass analysis to avoid perturbation of the negative ion signal by the electrons. An effusive beam of coronene is produced by vaporizing commercial solid samples (Merck) around 180 °C, in a double stage oven using a needle on top to allow a better definition of the collision centre. The electron and mass spectrometers are heated around 160 °C to avoid insulating deposits which strongly perturb low energy electrons, and preclude any reliable experiment after a few hours of vaporisation.

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3. Computational details

Density functional theory (DFT) calculations have been carried out using the hybrid B3LYP functional, which combines Becke's three parameter nonlocal hybrid exchange potential¹¹ with the non-local correlation functional of Lee, Yang and Parr,¹² and using the 6-311++G(d,p) basis set. All the calculations described here have been performed by means of the Gaussian03 suite of programs.¹³

To compute electronic excitations, we first optimized the geometry of the neutral coronene $C_{24}H_{12}$ (D_{6h} symmetry) at the B3LYP/6-311++G(d,p) level, and from the obtained structure vertical excitation energies and oscillator strengths were computed with the time dependent density functional theory^{14–16} (TDDFT) using the same functional and basis: TD-B3LYP/6-311++G(d,p). It has been previously shown that TDDFT is capable of reproducing low vertical excitation energies of PAH in good agreement with experimental results.^{17–23} This methodology, however, fails in the description of excitation energies higher than a certain threshold due to the wrong asymptotic behavior showed by the functionals, decaying faster than $1/R$ for large distances R from the nuclei. The threshold value is approximately the negative of the highest occupied Kohn–Sham orbital.¹⁴ We are interested in the lowest excitations and the results we present here (Tables 1–3 and Fig. 2) have a maximum energy of about 7 eV. As a benchmark to check the validity of the method, functional and basis, vertical excitation energies have been computed in benzene (C_6H_6) at the same level. The results obtained are in good agreement with experimental measurements.²⁴

The frontier orbitals of coronene as obtained with DFT (Kohn–Sham orbitals) are represented in Fig. 1. We show the highest occupied molecular orbitals (HOMO-1 and HOMO) and the lowest unoccupied molecular orbitals (LUMO and LUMO + 1). The four orbitals are doubly degenerate with e_{1g} (HOMO-1, LUMO) and e_{2u} (HOMO, LUMO + 1) symmetry. The first electronic excitations mainly concern these orbitals. We present in Tables 1 and 3 the results of the calculated singlet and triplet vertical excitation energies, in comparison with present experimental results and other previous works. Dipole allowed transitions to $^1E_{1u}$ states are presented separately for clarity in Table 2. Our results for the first singlet states (Table 1) are in nice agreement with previous TDDFT calculations.²⁰ However, the larger basis that we have employed in our calculations causes the appearance of additional states between 5.0 and 5.5 eV. In fact, it has been recently shown that including diffuse basis functions is of great importance to account for Rydberg or diffuse excited states;¹⁷ these states may appear among the low-lying excited states. The agreement obtained between our theoretical calculations and the present experimental measurements for dipole allowed transitions, even for energies up to 7.0 eV, gives confidence in the theoretical approach here proposed (see results in Table 2 and Fig. 2).

To help the understanding of the present vibrational excitation experimental results, vibrational harmonic frequencies have been also computed using the same level of theory B3LYP/6-311++G(d,p). Our results are in good agreement

with previous studies.^{25–29} They are not presented in this article as they do not show any significant improvement with respect to the already published values.^{25–29} However, we have used them to determine the nature of the excited vibrational modes observed in the experiment.

Vertical electron affinity (VEA) of coronene was also computed as the difference between the total energies of the neutral and anionic species at the optimized geometry of the neutral. We obtained a value for the VEA of 0.456 eV.

4. Results and discussion

4.1 Electronic excitation

The electron energy loss (EEL) spectrum of gaseous coronene has already been the object of several investigations, either at very high electron energy and low resolution,⁶ or intermediate energy and a resolution of about 0.1 eV.^{7–9} The present work, carried out with a substantially improved energy resolution (0.025 eV FWHM), allows more detailed observations. TDDFT calculations have been undertaken to help in assignments of the observed bands.

It is well known³³ that EEL spectroscopy can reveal both dipole allowed transitions and spin or symmetry forbidden transitions, by variation of the incident electron energy and the scattering observation angle. Shortly, when the incident energy is high compared to the excitation energy ($E_i \gg E_{exc}$), the incident electron has to lose only a small amount of energy; the collision occurs therefore at rather large impact parameters and the scattered electron is observed mainly at very small scattering angles θ (a few degrees). In these conditions the exchange process is unlikely, and the spectrum reveals essentially the spin allowed transitions. On the contrary, when $E_i \approx E_{exc}$ the collision occurs at low impact parameters, the electron penetrates the molecular cloud, and the exchange process becomes likely. The scattered electron can be observed at various angles. At larger scattering angles, the intensity of transitions to singlet states, strongly peaked at small angles decreases rapidly, and the generally less intense spin forbidden transitions states can therefore be observed. Our spectra have been recorded in the constant residual energy mode ($E_r = E_i - E_{exc}$). This mode allows a better comparison between intensities of the different processes, as the energy above threshold is the same for the various processes involved. Furthermore, when the residual energy is only a few eV, this avoids any transmission discrimination in the electron analyzer.

The EEL spectrum showing the singlet states ($E_r = 80$ eV, $\theta = 4^\circ$) is presented on Fig. 2a. Main bands appear at 3.78; 3.94; 4.28; 4.45; 5.90 and 6.36 eV (all values ± 0.02 eV). Several shoulders, particularly around 5.4 eV and between 6.5 and 7.5 eV, are also observed due to weaker transitions. The main features of our spectrum are in good agreement with less-detailed previous EEL spectra in the gas phase,^{7,9} and with photo-absorption data.^{4,34} Note that in our experiment, carried out at intermediate incident energy, the intensity of the band at 6.36 eV is underestimated as the condition $E_i \gg E_{exc}$ is only approximate. This is also the case for the data obtained at similar intermediate energy

Table 1 Vertical excitation energies (eV) and oscillator strengths (in parentheses) of the first singlet excited states of coronene. Theoretical results of this work (a), from ref. 20 (b) from ref. 51 (c). Experimental results of this work (d), from ref. 37 (e), from ref. 38 (f), from ref. 7 (g), from ref. 34 (h), from ref. 52 (i)

Singlet states	(a) This work TDDFT–B3LYP/6-311++G(d,p)	(b)TDDFT-RPA	(c) CNDO/S	(d) This work EEL Exp.	(e) Exp	(f) Exp	(g) Exp	(h) Exp	(i) Exp
1^1B_{2u}	3.20	3.20	3.12	2.94	3.03	2.90			2.90
1^1B_{1u}	3.41	3.42	3.56	3.77	3.65	3.63	3.69	3.74	3.63
1^1E_{1u}	4.09 (0.67)	4.12 (0.67)	4.34 (2.11)	4.28	4.09	4.06	4.20	4.25	4.07
1^1A_{2g}	4.23	4.32		4.45				4.42	
1^1E_{2g}	4.23	4.30	4.22						
1^1A_{1g}	4.34	4.40							
2^1A_{2g}	4.54	4.54	4.31						
2^1E_{2g}	4.61	4.63	4.65						
3^1E_{2g}	4.83	4.93							
2^1A_{1g}	4.97	5.03							
1^1E_{2u}	5.01								
4^1E_{2g}	5.15	5.19							
1^1B_{1g}	5.17								
1^1E_{1g}	5.20								
1^1B_{2g}	5.23			5.38					
5^1E_{2g}	5.37	5.38							
2^1B_{2u}	5.38	5.45		5.52					
1^1A_{2u}	5.53 (0.01)								
2^1E_{2u}	5.54								
1^1A_{1u}	5.54								
2^1B_{1u}	5.67								
2^1E_{1u}	5.81 (0.00)								
6^1E_{2g}	5.92								
3^1E_{2u}	5.93								
3^1E_{1u}	5.94 (0.28)			5.9			5.85	5.32	
2^1E_{1g}	5.97								
3^1E_{1g}	6.00								
4^1E_{1u}	6.08 (0.02)								
2^1B_{2g}	6.17								
2^1B_{1g}	6.20								
4^1E_{1g}	6.20								
5^1E_{1u}	6.21 (0.01)								
6^1E_{1u}	6.29 (0.90)			6.37			6.32	6.19	
5^1E_{1g}	6.30								
3^1B_{2u}	6.37								
7^1E_{1u}	6.44 (0.29)								

Table 2 Vertical excitation energies (eV) and oscillator strengths (in parentheses) of the first dipole allowed excited states of coronene (1^1E_{1u} symmetry) with an oscillator strength larger than 0.05 calculated at the TDDFT–B3LYP/6-311++G(d,p) level of theory in comparison with the experimental values extracted in the electron energy loss measurements; (s) indicates a shoulder in the spectrum

State	Calc. (this work)	$f > 0.05$	Exp. (this work)
1^1E_{1u}	4.09	(0.67)	4.28
1^1E_{1u}	5.94	(0.28)	5.9
1^1E_{1u}	6.29	(0.90)	6.37
1^1E_{1u}	6.44	(0.29)	6.7 (s)
1^1E_{1u}	6.75	(0.41)	6.8 (s)
1^1E_{1u}	6.80	(0.14)	7.0 (s)

($E_i = 100$ eV) by Delfs *et al.*,⁹ but curiously not for Khakoo *et al.*,⁷ also recorded at $E_i = 100$ eV, where the band at 6.36 eV seems overestimated. Experiments carried out at very high incident energy do not suffer this limitation.^{6,8}

The angular behaviour $I = f(\theta)$ of the three main transitions to singlet states, normalized at $\theta = 4^\circ$, is displayed on Fig. 3, showing the strongly peaked behaviour at very low scattering angles. The transitions at 3.78 and 4.28 eV behave identically whereas the band at 6.36 eV has a slightly different decrease.

This could possibly indicate some contribution of a plasmon structure, superimposed to the single electron transitions, as already observed for fullerenes.^{35,36}

Our computed excitation energies and oscillator strengths of dipole allowed transitions are in good agreement with the experimental data (Fig. 2b and Tables 1 and 2). It could appear surprising that below 4 eV, despite clearly observed peaks, all calculated transitions present a zero oscillator strength. Indeed, due to the high symmetry of coronene (D_{6h}), dipole allowed transitions are only to 1^1E_{1u} and 1^1A_{2u} states. The first band clearly observed at 3.78 eV is due to a dipole forbidden transition to the second singlet state S_2 (1^1B_{1u}). This transition is however observable, even in photo-absorption experiments in solution,^{37,38} due to vibronic coupling through e_g vibrations to the first 1^1E_{1u} state appearing at 4.28 eV.

Computed excitation energies and oscillator strengths f of dipole allowed transitions are directly compared with experiment on Fig. 2b. The first allowed transition to the 1^1E_{1u} state is computed only 0.190 eV lower than the observed value. An even better fit is observed around 6 eV for several dipole allowed transitions. However, this very good agreement

Table 3 Vertical excitation energies (eV) of the first triplet excited states of coronene. (a) Calculated values of this work at the TDDFT–B3LYP/6-311++G(d,p) level; (b) experimental results of this work from electron energy loss experiments; (c) experimental results from ref. 37; (d) experimental results from ref. 38

Triplet states	(a) This work TDDFT–B3LYP/6-311++G(d,p)	(b) This work EEL Exp.	(c) Exp	(d) Exp
1^3B_{1u}	2.35	2.40	2.40	2.37
1^3E_{1u}	2.95			
1^3B_{2u}	3.08			
1^3E_{2g}	3.57			
1^3A_{2g}	3.72			
1^3A_{1g}	3.92			

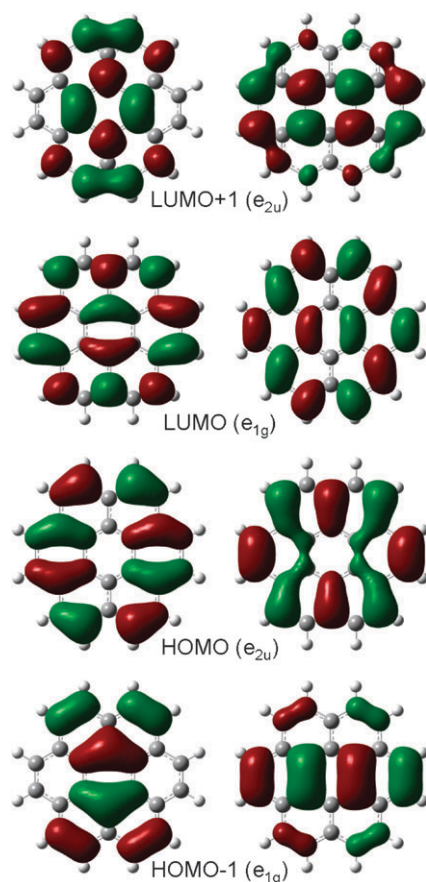


Fig. 1 Shape of frontier orbitals of coronene: highest occupied molecular orbitals (HOMO-1 and HOMO) and lowest unoccupied molecular orbitals (LUMO and LUMO + 1) obtained at the B3LYP/6-311++G(d,p) level of theory. The same isosurface has been used in all the plots.

should not be emphasized as the TDDFT method may not be totally reliable at these rather high excitation energies.¹⁷ As already mentioned, the computed f values cannot be directly compared to our observations at intermediate incident energy, intensities of transitions around 6 eV being particularly experimentally underestimated. The symmetry forbidden transition to the S_2 state, presenting a zero oscillator strength, does not appear on Fig. 2b.

The EEL spectrum at lower residual energy $E_r = 3$ eV and larger scattering angle $\theta = 15^\circ$ is displayed on Fig. 4. In these conditions, transitions to the singlet states 1^1B_{2u} and 1^1E_{1u} are

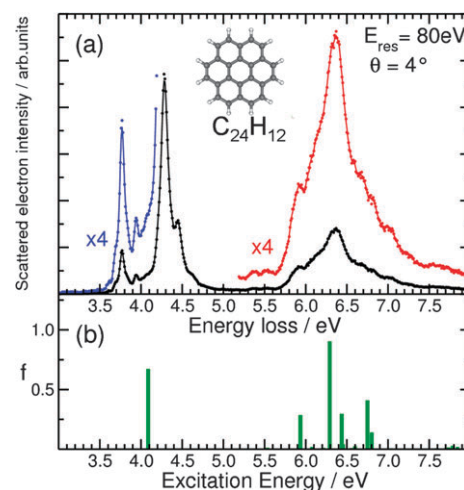


Fig. 2 (a) Experimental electron energy loss spectrum of coronene recorded with a residual energy of $E_{res} = 80$ eV and a scattering angle of $\theta = 4^\circ$. (b) Excitation spectrum at the TDDFT [B3LYP/6-311++G(d,p)] level: oscillator strength [f] as a function of the excitation energy for the dipole allowed transitions (bars).

strongly reduced but still present. Two other bands, clearly observed at lower energy losses originating at 2.41 and 2.95 eV (all values ± 0.02 eV), are attributed respectively to 1^3B_{1u} and to the first singlet state S_1 (1^1B_{2u}). The observation of S_1 , a symmetry forbidden state, extremely weak at $E_r = 80$ eV is easily observed at $E_r = 3$ eV. Note that this state can also be observed in photo-absorption experiments in solution (but not in the gas phase see ref. 4), due to a vibronic coupling to E_{1u} states through e_g vibrational modes (like for S_2).^{37,38} The vibrational structure of S_1 shows peaks at 2.95, 3.01, 3.11 3.20, and 3.31 eV. The computed values for excitation of the lowest triplet states and of S_1 and S_2 are also shown by arrows on Fig. 4. Our computed value for S_1 at 3.20 eV (Table 1) is in good agreement with the experiment, whereas the value for S_2 at 3.41 eV appears lower than the observed peak (3.78 eV). Although obtained with a lower resolution, our results for the vibrational structure are in good agreement with photo-absorption data.^{37,38} At low electron residual energy the contribution of higher triplet states (Table 3) could also occur in the S_1 energy region. The similarity of our results with the photo-absorption data is an indication that the contribution of these higher triplet states, computed at 2.95 and 3.08 eV, is weak.

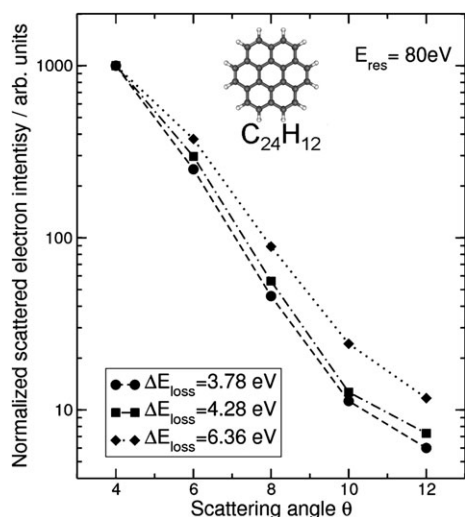


Fig. 3 $I = f(\theta)$ of the three main transitions to singlet states. Normalized scattered electron intensity as a function of the scattering angle is shown for the transitions at energies 3.78, 4.28 and 6.36 eV.

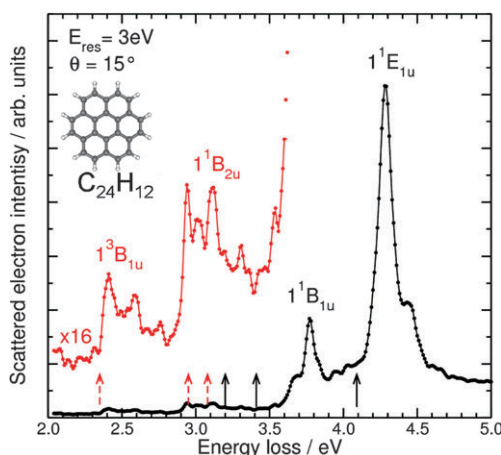


Fig. 4 Electron energy loss spectrum of coronene recorded with a residual energy of $E_{\text{res}} = 3$ eV and a scattering angle of $\theta = 15^\circ$. Arrows show the position of the first triplet states (red dashed lines) and the first singlet states (black full lines) as calculated at the TDDFT-B3LYP/6-311++G(d,p) level.

The first triplet state 1^3B_{1u} is observed at 2.41 eV, in excellent agreement with previous data.^{37,38} Other peaks at 2.59 and 2.77 eV reveal a vibrational structure with spacings of about 0.180 ± 0.01 eV. Again, our computed value of 2.35 eV for this state (Table 3) shows a very good fit with the experiment.

4.2 Vibrational excitation

At impact energies below 10 eV, the interaction of an electron with a molecule is dominated by the existence of very short lived anions (“resonances”), due to the temporary capture of the incident electron by the target. The lifetime of these resonances (10^{-10} – 10^{-15} s) does not generally allow

their direct observation. They are observed through their decay, either by autodetachment or by dissociation. If the lifetime of the resonance is not too short, the rejection of the electron occurs at an inter-nuclear distance different from the ground state, giving rise to vibrational excitation. This decay channel is in competition with the dissociation channel which leads to a stable negative ion and one or several neutral fragments, *via* the dissociative electron attachment process (DEA).

It has been shown that resonant vibrational excitation of molecules of high symmetry can be very selective.^{39–41} Indeed in the case of benzene,³⁹ among the 20 different frequencies of the 30 vibrational modes of the molecule, the vibrational excitation through the first resonance $^2\text{E}_{2u}$ at 1.1 eV concerns only a very few modes (ν_2 , ν_{16} , ν_{20}). Furthermore, a few eV above the first resonance, the second resonance $^2\text{B}_{2g}$ at 4.8 eV gives rise to the excitation of only 2 different modes (ν_1 , ν_7). As coronene and benzene belong to the same symmetry group D_{6h} , some selectivity for vibrational excitation in coronene could also be expected despite the complexity of the molecule (102 modes, 68 different frequencies). To our knowledge there is only one previous work on vibrational excitation of gas phase coronene,⁷ performed with an electron energy resolution of about 0.08 eV FWHM. Their EEL spectrum ($E_i = 11$ eV $\theta = 30^\circ$) indicates the excitation of the CH stretch modes; a wider bump around 0.190 eV lets us suspect some other unresolved excitations.

In the present work vibrational excitation has been studied at incident energies varying from 1.5 to 9 eV with an energy resolution of 0.025 eV (FWHM, energy loss mode). The EEL spectra at $E_i = 1.5$ to 9 eV recorded at $\theta = 90^\circ$ are displayed on Fig. 5. All spectra have been normalized to allow a comparison of vibrational excitations intensities

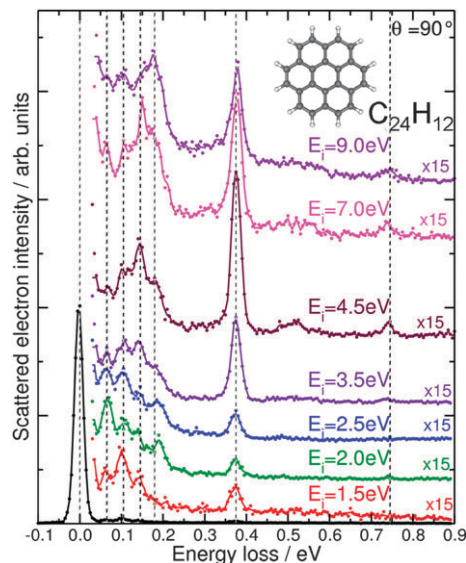


Fig. 5 Electron energy loss spectra of coronene recorded for different values of incident energy ($E_i = 1.5, 2.0, 2.5, 3.5, 4.5, 7.0$ and 9.0 eV) recorded at a scattering angle of $\theta = 90^\circ$. All spectra have been normalized to the elastic peak. Dashed guidelines are placed at 0.0, 0.065, 0.105, 0.145, 0.180, 0.375 and 0.745 eV.

at different energies. Even if our improved electron energy resolution obviously does not allow us to identify each of the different vibrational excited modes, rather well separated contributions are observed at energy losses between 0.060 and 0.200 eV, besides the CH stretch modes appearing at 0.375 eV. Note that the excitation of infrared active modes by a direct process is usually essentially observed at very low incident energy and small scattering angle. IR spectroscopy shows⁴² that the most intense modes observed occur at 0.100 eV and 0.375 eV with similar intensities. Medium contributions are found at 0.068 eV and 0.165 eV. Weaker bands occur at 0.140 eV and 0.185–0.195 eV. In our EEL spectrum at $E_i = 1.5$ eV and $\theta = 90^\circ$ the band at 0.165 eV does not appear at all, and the intensity ratio between the bands at 0.065 and 0.145 eV does not reflect the IR results. We are therefore not dealing with direct excitation process. However, despite the large observation angle (90°), a weak contribution of IR modes excited by a direct process cannot totally be excluded at this rather low incident energy. At higher incident energies, the variation of the relative intensities of the vibrational bands is indicating the occurrence of several resonant excitation processes. We can roughly distinguish 4 different types of EEL spectra (a) around 1.5 eV, (b) from about 2 to 3.5 eV, (c) from about 4 to 7 eV and (d) around 9 eV to 11 eV (the spectra at 9 and 11 eV (not shown) are similar). Electron transmission spectroscopy (ETS), a particularly adapted technique to give the energy position of resonances, has been carried out on numerous conjugated aromatic molecules from benzene to tetracene⁴³ but unfortunately not for coronene. It shows that the number of π^* resonances observed up to about 5 eV is increasing with the size of the molecule. The possible existence of several π^* resonant states between 1 and 5 eV in coronene is therefore not surprising. The electron affinity of coronene being positive,³⁰ the first anion state (π^*_1) having one extra electron in the LUMO $\dots(e_{2u})^4(e_{1g})^1{}^2E_{1g}$ is stable.

Only a very few studies are devoted to the excited states of the coronene anion. Experimentally, an absorption spectrum in solution with THF and alkali metals⁴⁴ recorded up to 5 eV, has shown several very weak excitations between 0.5 and 1.7 eV followed by four intense bands, rather sharp, peaking at about 2.0; 2.7; 3.6 and 4.0 eV. In a recent theoretical work, Mallocci *et al.*³² have obtained an excitation spectrum of the coronene anion. They found very small excitations around 1.5; 2 and 3 eV, followed by 2 intense bands at about 3.4 and 5.8 eV. These values are not inconsistent with the experimental observations which extend only up to 5 eV.

The excited bands observed in the photo-absorption spectrum give an indication of the resonant states which can contribute to vibrational excitation. However, some other resonant states, not dipole allowed, might also exist, and among the observed states, some could correspond to core excited resonances not coupled to the ground state. Therefore, it is not easy to know precisely the number of low lying resonances involved in the excitation process. These resonances will certainly overlap making the modifications of the vibrational excited modes only gradual when the incident energy is raised.

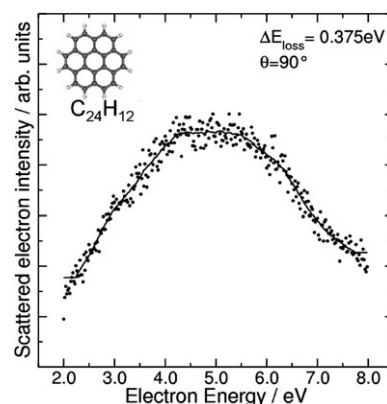


Fig. 6 Excitation function for the ν_{CH} stretch mode, scattered electron intensity as a function of the electron energy for $\Delta E_{loss} = 0.375$ eV and $\theta = 90^\circ$.

Excitation functions of these modes could in principle reveal the different resonant states involved. However, except for the CH stretch modes, the various groups of vibrational excited modes are not separated enough to allow recordings of reliable excitation functions for each group. The excitation function of the CH stretch modes is presented in Fig. 6. It shows a very smooth behaviour, with a maximum around 5 eV, which does not reveal the various expected π^* resonances. However, the ν_{CH} stretch modes are not supposed to be excited through π^* resonances but rather through a σ^* resonance. This σ^* resonance is wide and appear to cover most of the energy range studied from 2 to 8 eV.

To be compared with our observations, the energy position of the bands found in the photo-absorption spectrum of the anion has to be subtracted from the electron affinity of coronene (0.47 eV³⁰). This gives the expected resonances corresponding to the most intense bands, at about 1.5 eV; 2 eV; 3.1 eV and 3.5 eV. These values are rather consistent with our observations.

At $E_i = 1.5$ eV (π^*_2 region) besides a weak ν_{CH} excitation, 3 groups of modes are excited at energy losses 0.065; 0.105 and 0.145 eV. At $E_i = 2.0$ eV and 2.5 eV (π^*_3 region), the 0.065 eV band is more strongly excited and another band at 0.190 eV is observed which reduces quickly above 2.5 eV. At $E_i = 3.5$ eV this last band becomes only a shoulder, whereas the excitation of CH stretch modes appears dominant. From 3.5 to 7 eV (π^*_4 region and σ^* region) the excitations around 0.065; 0.105 and 0.180 eV vanish progressively to the benefit of the band at 0.145 eV; the CH stretch becomes dominant, 2 ν_{CH} and a combination band $\nu_{CH} + 0.145$ eV being also observed. This indicates clearly excitations *via* a σ^* resonance. At $E_i = 9$ eV and above, the CH excitation decreases, whereas the group of modes at 0.105 eV and 0.175 eV reappears.

As already mentioned, assignment of separate vibrational modes is clearly out of reach with the present resolution (8 ν_{CH} modes are ranging from 0.375 to 0.378 eV⁴⁵). However, with the help of our DFT calculations, we have been able to characterize the main nature of the various vibrational modes excited in the energy loss region between 0.060 to 0.200 eV as described below.

—around 0.065 eV the excited vibrational modes correspond mainly to CCC skeleton deformations out of plane.

—around 0.105 eV the excitation concerns essentially CCH bending modes out of plane and also some contribution of CC stretch modes in plane.

—around 0.145 eV the excitation is due to CCH bending modes in plane, with a small contribution of CC bending modes in plane

—in the region 0.180–0.190 eV the main modes involved are due to CC stretch modes in plane.

The first anion state π^*_1 (${}^2E_{1g}$) lying below the neutral state is not a resonance. The first resonant state π^*_2 is likely to have the extra electron in the LUMO + 1 (${}^2E_{2u}$). From the sketch of the LUMO + 1 (Fig. 1), it can be seen that the CCC modes will be strongly affected. In the resonance region 2–3 eV, this could explain the appearance of the band around 0.185 eV (CC stretch modes), and due to the π^* nature of this orbital, the observation of a clear excitation of CCC deformation modes out of plane at 0.065 eV.

Above 3.5 eV the contribution of the σ^* resonance becomes important and is mixed with the higher π^* resonances; the dominant excitations concern CH stretch modes at 0.375 eV and also the CCH bending modes in plane at 0.145 eV.

Above 7 eV several resonant states are likely to strongly overlap giving the analysis ambiguous.

4.3 Negative ions

The electron affinity (EA) of coronene is known to be positive. Our DFT calculations lead to 0.456 eV for the vertical electron affinity (VEA). This value is in excellent agreement with previous experimental measurements: 0.54 ± 0.1 eV using a kinetic method,³¹ and 0.47 ± 0.09 eV in a photo-detachment experiment.³⁰ Our result is also in agreement with recent theoretical predictions by Mallocci *et al.*,³² and by Denifl *et al.*⁵ (see ref. 5 for an extended discussion on electron affinity of coronene).

Formation of negative ions in coronene by electron impact has already been investigated recently.⁵ These authors have found only 2 anions *i.e.* the parent anion $M^-(C_{24}H_{12})^-$ essentially produced at zero energy, and the dehydrogenated anion $(M-H)^-$ formed mainly around 8 eV, and also at 0.1 eV with a very small intensity. Note that formation of parent anions by electron impact around zero energy is a rather common process for large molecules.⁴⁶ The most surprising case occurs for fullerenes where long lived parent anions ($\tau > 1$ μ s) have been observed up to 15 eV.^{47–49} In PAHs, Tobita *et al.*⁵⁰ have shown that a long lived parent anion is generally observed when the electron affinity (EA) of the PAH is positive, and not observed if the EA is negative. However, even when the EA is positive, the zero energy peak is sometimes missing, whereas long lived parent anions can be observed at slightly higher energies. The present results, in agreement with Denifl *et al.*,⁵ confirms that the coronene parent anion is the major anion produced at zero energy. The variation of this anion yield *versus* electron energy is displayed on Fig. 7. The better electron energy resolution allows to demonstrate the existence of 2 peaks, in contrast

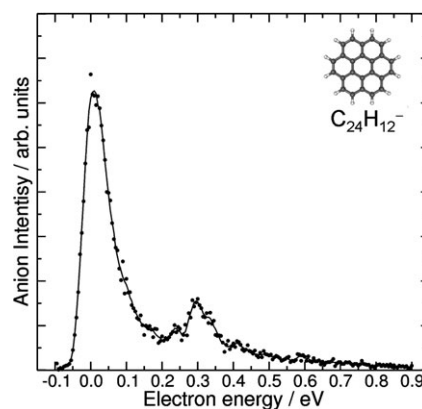


Fig. 7 Anion yield *versus* incident electron energy.

with previous data. The main peak occurs at zero energy (within 0.03 eV), and a second weaker peak appears at 0.3 eV. Note that due to the decrease of the incident electron beam intensity close to zero energy, the intensity of the peak at zero is likely to be underestimated. The nature of this second peak is not obvious. Such peaks at energy higher than zero have also been observed in other PAH by Tobita *et al.*⁵⁰ In tetracene for example, they showed that the energy position of the higher energy peaks was corresponding to the resonant states revealed by ETS.⁴³ As already mentioned above, some indication of the anion excited states in coronene can be found in the photo-absorption spectrum of the anion in solution.⁴⁴ This spectrum shows that the first excited states appear only at about 0.5 eV. However, lower anion states, not dipole allowed, can also exist at lower energy. Indeed, the comparison of the resonant states found in ETS and the bands found in absorption spectra of various anions shows actually that the first resonances lie generally below the first band.⁴³ We have indicated above (Tobita *et al.*⁵⁰) that parent anions corresponding to negative EA are generally not observed due to the auto-detachment towards the ground state. The peak at 0.3 eV can therefore correspond, either to a particularly long lived shape resonance, or to a resonant state having a core excited character, not directly coupled to the ground state of the anion. As the spacing between the two peaks is reminiscent of the CH stretch mode (0.375 eV), a vibrational Feshbach resonance associated with this mode can also be involved. Electron transmission experiments would certainly be very useful to identify the resonant features close to zero energy.

5. Conclusion

By varying the incident energy and the scattering observation angle, EEL spectra in gas phase coronene have shown both dipole allowed and symmetry or spin forbidden transitions up to 8 eV. For allowed transitions to singlet states, TDDFT excitation energies and oscillator strength are in rather good agreement (within about 0.2 eV) with experimental data, even up to 7 eV excitation energy. The energy of symmetry forbidden transitions correctly reproduced for the S_1 (${}^1B_{2u}$) state, appears slightly too low for S_2 (${}^1B_{1u}$). For the lowest

triplet state ($^3B_{1u}$) the agreement is excellent. Even with the lower energy resolution of our electron impact data, the vibrational structure of the S_1 band agrees quite well with photo-absorption data.

Detailed study of pure vibrational excitation is presented for the first time. Like in benzene, it shows some selectivity, 5 groups of vibrational modes being clearly differently excited when the incident energy is varied. This indicates the presence of several π^* resonant states and one wide σ^* resonance. Electron transmission experiments would clearly help to precise the exact number of resonances existing in the energy range 0 to 9 eV.

Concerning the coronene anion, our calculated electron affinity (0.45 eV) agrees well with other calculated or measured values. We confirm previous data indicating that the only anion observed at zero energy is the parent anion ($C_{24}H_{12}^-$). However, in contrast with this data, besides the major peak at zero, we have found another peak at 0.30 eV (all values within 0.030 eV) in the anion yield versus electron energy. This peak can possibly be due to a core excited anion state, not directly coupled to the ground state or to a vibrational Feshbach resonance associated with the CH stretch mode.

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